Questions: The Creative Agent

Analytical Questions

1. Define the inventor as an Active Inference agent. What are the key components (generative model, prior beliefs, precision weighting, action policies) and how do they interact to produce inventive behavior?

2. Compare and contrast the Methodical Engineer and the Tinkerer archetypes in terms of their Active Inference parameters. How do differences in prior strength, precision weighting, and exploration rate lead to different inventive strategies?

3. Explain the concept of "functional fixedness" as an Active Inference phenomenon. How do overly precise priors about object categories prevent inventors from seeing novel uses, and what does the framework suggest about how to overcome it?

4. How does the expected free energy decomposition (pragmatic + epistemic value) illuminate different types of inventor motivation? Why might a purely pragmatically-motivated team produce different inventions than one with strong epistemic motivation?

5. Analyze team creativity as multi-agent Active Inference. What is the role of complementary generative models, and why is model diversity valuable up to a point but potentially dysfunctional beyond it?

6. Steven Sasson, a relatively new engineer at Kodak, invented the digital camera in 1975, while Kodak's senior photography experts missed the opportunity. Explain this using the concept of prior beliefs and precision weighting. Why was being new an advantage?

7. Active Inference predicts that agents select policies minimizing expected free energy. How does this apply to an inventor choosing between refining an existing prototype (exploitation) and trying a radically different approach (exploration)? What factors tip the balance?

8. How do shared artifacts (prototypes, sketches, diagrams) function in multi-agent creative systems? Explain their role as shared sensory evidence that multiple generative models can process, using the concepts from this module.

9. The "accidental discoverer" archetype requires both strong domain expertise (to have a model that can be violated) and openness to anomaly (to notice when it is violated). Analyze this apparent paradox. How can an inventor cultivate both traits simultaneously?

10. Consider the concept of "precision weighting" in creative teams. When two team members disagree about a design decision, how might their disagreement be understood as a difference in precision weighting rather than a difference in preferences? What are the implications for resolving creative conflicts?

Applied Questions

1. Describe your own generative model as an inventor. What are your core beliefs about the problem you are solving, and how confident are you in each? Identify at least one belief you hold with confidence that might be wrong.

2. Using the archetype framework from this module, classify yourself. What is your primary archetype, and what evidence from your past creative work supports this classification? How has your archetype shaped the inventions you have pursued?

3. Identify three specific blind spots created by your inventor archetype. For each, describe a situation where this blind spot could lead you to miss an important aspect of your invention project.

4. Choose an inventor archetype that is opposite to yours. Spend ten minutes thinking about your invention idea from that perspective. What new possibilities emerge? Document at least two ideas that you would not have generated from your default perspective.

5. Design a creative team of 3-4 people for your invention project, specifying each member's archetype and the unique generative model they bring. Explain how the team's model diversity covers blind spots that no individual member could address.

6. Think of a creative disagreement you have had with a collaborator. Reanalyze that disagreement using the Active Inference concepts from this module. Were you disagreeing about facts, or about which prediction errors to prioritize (i.e., different precision weightings)?

7. Percy Spencer noticed that a chocolate bar melted near a magnetron and invented the microwave oven. Describe a time when you noticed something unexpected in your own work. Did you investigate it or dismiss it? What does your response tell you about your agent configuration?

8. The module discusses the tension between shared priors (necessary for communication) and diverse priors (necessary for creativity) in teams. Describe a specific strategy you would use to manage this tension in your own inventive practice.

9. Assess your motivational profile: what proportion of your creative drive is pragmatic (solving a specific problem, achieving a specific outcome) versus epistemic (satisfying curiosity, understanding how things work)? How does this profile affect the types of projects you choose and the way you approach them?

10. Write a "Creative Agent Specification" for yourself: document your key priors, precision weightings, preferred action policies, motivational balance, and known blind spots. Include a concrete plan for addressing your most significant blind spot on your current invention project.